PIPPA

Raw ATAC-seq sample reads

Stage 1: Confirm quality of ATAC-sequencing data

Raw reads QC
- Sequences quality
- GC content
- Duplication
- Length distribution

Pre-alignment QC
- Removal of low-quality bases
- Reads trimming
- Removal of adapters

Alignment
- Reads aligned to reference genome
- Removal of reads improperly paired or of low mapping quality.
- Removal of duplicated reads.

Post-alignment QC
- ENCODE metrics
- Fragment size distribution
- TSS enrichment score

Peak calling
- Peak calling at sample level
- FRiP calculation

Aligned bam sample

Stage 2: Confirm the correct individual

Concordant sex prediction
- Peak calling on sex chromosomes.
- Counts in peaks for sample classification

Concordant genetic information
- Compare genetic with external SNP data.
- Identify potential sample swaps.

Stage 3: Confirm the correct cell type

Peak calling
- Group samples by labelled cell type
- Peak calling at group level with Q < 0.05.

Cell type clustering
- Counts in peaks for samples classification
- Normalisation of counts
- Check concordant cell type level based on clustering

High-quality ATAC-seq cell type specific samples

Peak set curation for downstream analysis

Initial peak calling
Discovery peak calling on merged bam samples file with Q < 0.05

Threshold discovery
Plot peak numbers versus significance threshold from Q < 0.05 to Q < 1e-100

Threshold peak calling
Peak calling on merged samples bam at indicated threshold

Peak set verification
- Peak number
- Width of peaks
- Inter-peak distance
- Genomic annotation
- FRiP calculation

Read counts in cell type curated peak set